ETHICAL HACKING

VAPT LAB

FINAL PRACTICAL PROJECT

Vulnerability Assessment & Penetration Testing
From network identification to remediation and retest

```
$ nmap -sV -p 8000 127.0.0.1
  8000/tcp open  SimpleHTTP/0.6
$ curl http://127.0.0.1:8000
  [!] .ssh/  .gnupg/  Documents/ ...
$ _
```

A small Python server exposed private folders. What went wrong? How was it fixed?

Rohan Yethekar

Kali Linux | Authorized local lab
Scope: 127.0.0.1:8000

VAPT Lab Report

Vulnerability Assessment & Penetration Testing on Kali Linux

Student	Rohan Yethekar
Environment	Kali Linux - authorized local lab
Scope	127.0.0.1:8000 (controlled Python HTTP service)
Result	Finding identified, remediated and verified by retest

Workflow followed

Authorized testing only. All documented testing was performed against the student's own controlled lab environment. No unauthorized target was tested.

1. Objective

- Perform a controlled VAPT workflow from network identification through reconnaissance, service enumeration, HTTP analysis, remediation and retesting.
- Build practical skills with Linux networking, Nmap, curl and security evidence.

2. Network Identification

Item	Value
Interface	eth0
IPv4	10.0.2.15/24
MAC	08:00:27:8a:35:d2
Gateway	10.0.2.2
Network	10.0.2.0/24
Loopback	127.0.0.1

3. Connectivity

- `ping -c 4 10.0.2.2` - 4/4 received, 0% packet loss.
- `ping -c 4 8.8.8.8` - 4/4 received, 0% packet loss.

4. Port Discovery

- `ss -tuln` was used to observe listening sockets.
- A controlled Python HTTP service was started on TCP port 8000.

5. Nmap Enumeration

- `nmap -p 8000 127.0.0.1` - 8000/tcp open.
- `nmap -sV -p 8000 127.0.0.1` - SimpleHTTP/0.6 (Python 3.13.12).

6. HTTP Header Analysis

- `curl -I http://127.0.0.1:8000` - HTTP/1.0 200 OK.
- Server header exposed SimpleHTTP/0.6 Python/3.13.12.
- This is an information-disclosure observation, not proof of a critical vulnerability.

7. Finding: Directory Listing

- Initially, `curl http://127.0.0.1:8000` showed a directory listing from the Kali user's home directory.
- The listing included entries such as .ssh/, .gnupg/, Documents/ and Downloads/.
- Risk depends on exposure, permissions and the data being served.

8. Remediation

- Stopped the original server.
- Created `~/VAPT_Lab/web` with a controlled index.html: VAPT Lab - Authorized Security Testing.
- Restarted with `python3 -m http.server 8000 --bind 127.0.0.1`.

9. Retest

- `curl http://127.0.0.1:8000` returned only the controlled lab message.
- `nmap -sV -p 8000 127.0.0.1` confirmed the intended HTTP service remained available.
- Remediation was successfully verified in the lab.

10. Findings Summary

Finding	Risk / recommendation
Open HTTP service	Context-dependent exposure risk; restrict unnecessary services.
Server banner disclosure	Low/context-dependent; minimize unnecessary version disclosure where appropriate.
Directory listing / excessive file exposure	Configuration-dependent; use a dedicated web directory and restrict exposure.

11. Conclusion

- Completed workflow: network identification, connectivity, port discovery, service detection, HTTP analysis, finding, remediation, retest.
- No unauthorized target was tested.

12. Interview Explanation

I performed a controlled VAPT lab on Kali Linux, identified an HTTP service with Nmap, analyzed headers and directory exposure, applied remediation using a dedicated directory and localhost binding, and verified the fix with a retest.